

ORCA — Build, Gap-Closure & Differentiation Report for SIH26176 (Team OCEAN-IQ)

Marine EcOsystem Reasoning with Collaborative Agents · Theme: Disaster Management · Category: Software · Prepared 8 September 2026

TL;DR

ORCA's architecture is technically credible but under-specifies exactly the things an INCOIS/NIOT/ISRO jury will probe — real data-ingestion mechanics, proactive push alerting, IMBL/marine-protected-area geofencing math, causal "why-did-productivity-decline" reasoning, and forecast-trust quantification. Every one of these gaps is closable within the hackathon using **currently-live, real endpoints** (INCOIS ERDDAP, IMD's JSON/GeoJSON APIs at `api.imd.gov.in`, CMEMS `copernicusmarine` toolbox, MOSDAC's `mdapi` client, PostGIS + marineregions/WDPA polygons).

The single most defensible, non-obvious differentiator is a rigorous **Advisory Reliability + Uncertainty layer** (conformal prediction + CRPS/Brier skill scoring backtested against NIOT OMNI buoys), fused with **boat-capability-relative risk framing** and **deterministic, replayable agent provenance**. None of the incumbent Indian systems (SAMUDRA, GEMINI, mKRISHI, Fisher Friend, Jal Anveshak) do any of these — they deliver forecasts; ORCA would *quantify how much to trust them and reason about why*.

Do NOT rebuild what already exists. INCOIS already ships PFZ advisories, 5–10-day Ocean State Forecast, the SAMUDRA app, GEMINI/NavIC alerts and the DAT-SG distress transmitter. ORCA must be explicitly positioned as the **agentic reasoning-and-trust layer on top of** these authoritative feeds, not a competing app. Say this on stage — it signals domain awareness and de-risks the "you're duplicating INCOIS" jury objection.

Key Findings

1. Gap Analysis Against the Problem Statement (clause by clause)

The official statement demands: (a) agentic autonomous planning + task decomposition; (b) multi-agent collaboration; (c) tool selection; (d) spatio-temporal reasoning; (e) explainable/evidence-based recommendations; (f) multilingual Indian regional languages; (g) multi-turn conversation + query refinement; (h) proactive safety alerts (cyclone, lightning, high waves); (i) geofencing near IMBL / restricted waters / MPAs; (j) route optimization + safe navigation; (k) causal questions ("why has fish productivity declined"); (l) the Disaster-Management theme; and stakeholder coverage across fishermen, researchers, coastal authorities, disaster agencies and maritime operators.

What ORCA already covers well: the agentic orchestrator (intent → planning → decomposition → tool selection → agent coordination → memory), the specialized-agent roster, the dynamic DAG, and the layered ingestion→orchestration→reasoning→evidence→trust→output stack. This is a solid skeleton and better-articulated than most competing teams will manage.

What is missing or under-specified — the jury will notice each of these:

#	Problem-statement clause	ORCA's current state	Gap to close
G1	Multilingual auto-detection & regional-language response	PPT lists multilingual support generically	Incumbent SAMUDRA already offers menu-based language switching in "English, Hindi, and Indian coastal languages (Gujarati, Marathi, Kannada, Malayalam, Tamil, Telugu, Odia and Bengali)" (per its Apple App Store listing). ORCA must go beyond menu switching to <i>automatic language + code-mixed detection</i> and <i>voice-first</i> interaction for low-literacy fishermen. Use Bhashini / AI4Bharat (IndicTrans2 NMT, IndicASR/IndicWav2Vec, IndicWhisper, Indic-Parler-TTS — mostly Apache-2.0).
G2	Multi-turn context & query refinement	"Memory" listed	Needs an explicit conversation-state schema, entity carry-over (location/time/vessel), and clarifying-question behaviour when the query is under-specified.
G3	Autonomous dataset discovery	"Tool selection" listed	Needs a tool/dataset registry with capability metadata so the planner can <i>choose</i> the right source (e.g. prefer INCOIS SST, fall back to CMEMS) — not hard-wired calls.
G4	Explainability & evidence provenance	"Explainable recommendation," "7-point integrity check"	The "7-point integrity check" and "Adversarial Reliability Nucleus" are vague marketing terms a scientific jury will challenge. Replace with a concrete provenance graph (which timestamp, which agent, which formula) and evidence-sufficiency gating with abstention .
G5	Proactive alerting vs pull queries	Architecture is pull/query-centric	The statement explicitly requires <i>proactive</i> cyclone/lightning/high-wave alerts. Needs a scheduled geofenced alert-evaluation loop that pushes warnings, not just answers questions.
G6	Geofencing incl. IMBL & MPAs	Listed as a USP but math unspecified	This is the highest-empathy real problem — the Sri Lankan Navy apprehended 346 Indian fishermen and seized 44 trawlers in 2025 (per Sri Lankan Navy statements reported in Indian press), overwhelmingly for crossing the IMBL in the Palk Strait. Needs concrete PostGIS distance-to-IMBL with predictive drift warning.
G7	Route optimization	"Safe Route" output	Needs a named algorithm and cost surface (see blueprint).
G8	Causal "why productivity declined"	"Causal Hypothesis Engine"	Must be grounded in real oceanographic drivers (SST anomaly, chlorophyll, upwelling, marine heatwave, freshwater flux, overfishing) — not a generic LLM guess.
G9	Disaster-Management	Weak	This is the <i>scored theme</i> . Integrate IMD RSMC cyclone bulletins, INCOIS tsunami/storm-surge/high-wave/swell alerts,

#	Problem-statement clause	ORCA's current state	Gap to close
	theme		NDMA/SACHET, and ideally the Common Alerting Protocol (CAP) XML standard for interoperable warnings. Add boat-recall / harbour-advisory / cyclone-shelter workflows.
G10	Stakeholders beyond fishermen	Fisherman-centric	Add explicit personas: a researcher view (raw data + provenance + export), a coastal-authority/disaster-agency view (aggregate risk map, fleet-at-sea count, alert dissemination), a maritime-operator view (port warnings, sea-area bulletins, route ETAs). Multi-persona is cheap to add and directly scores against the clause.
G11	Evaluation/jury dimensions	Not addressed	SIH idea evaluation is scored on <i>novelty, complexity, clarity, feasibility, practicability, sustainability, scale of impact, user experience, and future potential</i> Slideshare (per SIH 2025 guidelines); grand-finale jurors add <i>technical feasibility, research quality, team dynamics</i> . Build a slide for each.

Scientifically loose claims a domain jury (INCOIS/NIOT/ISRO scientists) will challenge — fix these in the PPT before the finale:

Treating PFZ as "where the fish are." PFZ is a *probabilistic aggregation indicator* derived from SST fronts + chlorophyll, issued **three times a week** (confirmed in the Jal Anveshak paper: "issued three times a week by INCOIS by incorporating the data on Sea Surface Temperature (SST) and Chlorophyll retrieved... from NOAA-AVHRR, Eumetsat Met-Op, Oceansat-II and MODIS Aqua"). It is not real-time and not a fish census. Frame ORCA's fishing intelligence as *"probability-of-aggregation, with confidence and age of advisory shown."*

Assuming Oceansat-3 provides operational SST. MOSDAC states the EOS-06 Sea Surface Temperature Monitor (SSTM) "developed a technical problem in scan mechanism and therefore, not in operation at present." Do not claim live Oceansat-3 SST; use INCOIS/CMEMS/NOAA SST and Oceansat-3 for **chlorophyll (OCM-3)** and **winds (SCAT-3)** only.

"Random Forest / Gradient Boosting" with no training set or validation plan. A jury will ask "trained on what labels, validated how?" Either specify the labeled dataset and a hindcast validation, or reclassify these as deterministic scoring rules.

Real-time satellite chlorophyll. Cloud cover and revisit gaps make ocean-colour intermittent; state that you gap-fill/composite and show data age.

Naive IMBL as an EEZ median line. The India–Sri Lanka maritime boundary is fixed by the **1974 and 1976 bilateral agreements**, not a computed median — use the agreed boundary geometry, not a generated one, or you will mis-warn fishermen.

2. Data Ingestion — How, Exactly

All mechanics below were verified against live endpoints/official docs. Build each source as an **adapter behind a common cache** so any one can fail without taking down the platform.

INCOIS — the primary Indian authority

INCOIS ERDDAP (<https://erddap.incois.gov.in/erddap>), server v2.30.0): **open, no key**. This is your best programmatic Indian source. Real dataset IDs confirmed live:

[incois_tmi_3day_datasets](#) — 3-day TMI SST (griddap)

[NOAA_AVHRR_AMSR_datasets](#) — blended daily OI SST (griddap)

[incois_quickscat_daily_datasets](#), [ascats_daily_datasets](#), [ascats_mnt_datasets](#) — scatterometer winds (griddap)

[incois_oceansat2_datasets](#) — Oceansat-2 OCM ocean colour / chlorophyll (griddap)

[Indian_ARGO_Floats](#) — ARGO profiles (tabledap)

griddap URL pattern: [\[.../griddap/<id>.nc?SST\[\(2026-09-01T00:00:00Z\)\]\[\(5\):1:\(25\)\]\[\(70\):1:\(95\)\]](#) (Bay-of-Bengal box; check dim order/names via the [.das](#) / Data Access Form). Filetypes: [.nc](#), [.csv](#), [.json](#), [.mat](#), [.geotif](#), [.transparentPng](#).

tabledap pattern: [\[.../tabledap/<id>.csv?var1,var2,time,latitude,longitude&time=>...&latitude=>5&latitude<=25&longitude=>70&longitude<=95\]](#)

Caveat: the exact griddap ID for the INCOIS **wave / Ocean-State-Forecast** product was not confirmable by name from the catalog snippets — verify in the browser catalog ([\[.../erddap/search/index.html\]](#)). Treat **CMEMS as the reliable wave source** (below).

INCOIS PFZ advisories — issued 3×/week; WebGIS at [incois.gov.in/MarineFisheries/PfzWebGis](#) and geoportal [incois.gov.in/geoportal/MFASPFZ](#); **text/bulletin data** at [incois.gov.in/MarineFisheries/TextDataHome](#). **No clean**

JSON API — you must parse the text advisories / WebGIS layers (WMS). Cache per coastal node (INCOIS publishes for ~1,223 coastline nodes).

INCOIS Ocean State Forecast (INDOFOS) — significant wave height, swell, currents, SST, mixed-layer depth, depth-of-20°C isotherm, astronomical tides, wind — **3-hour intervals, 5-10-day horizon**. Delivered via SAMUDRA app and bulletins; scrape/mirror OSF bulletins or use CMEMS equivalents.

INCOIS Algal Bloom Information Service (ABIS) — operational HAB (Noctiluca) detection service; a real anchor for ORCA's anomaly-detection module.

Scale context for the pitch: INCOIS reaches large fisher populations directly — e.g. on the order of ~100k-160k users per coastal state via WhatsApp/SMS/Telegram/SAMUDRA (PIB figures cite ~103,915 users in Andhra Pradesh and ~158,711 in Odisha). ORCA rides on top of this reach, it doesn't recreate it.

IMD — weather, cyclone, marine bulletins

<https://api.imd.gov.in/api/v1/...> — **public JSON, no data.gov.in key required** (IP-whitelisting optional for heavy use; IMD asks for attribution + client caching). Marine/disaster-relevant endpoints (all JSON, several GeoJSON):

[seabulletin?id=...](#) (Sea Area Bulletin — wind, sea condition, warnings), [coastalbulletin](#), [portwarning?id=...](#), [fishermen-warning](#)

[cyclone_track](#) (observed + forecast positions, MSW, category), [cyclone_wind](#) (**GeoJSON** 27/34/50/64-kt wind radii), [cyclone_cou](#) (**GeoJSON** cone of uncertainty)

[current_wx](#), [cityforecast](#), [districtnowcast](#), [stationnowcast](#), [districtwarning](#), [basingpf](#)

This is a **major win**: marine, cyclone, sea-area and coastal warnings are available as structured JSON/GeoJSON, so you avoid PDF scraping. RSMC New Delhi (the official Bay-of-Bengal/Arabian-Sea tropical-cyclone RSMC) also publishes National/RSMC/GMDSS/Fishermen-Warning bulletins.

ISRO / MOSDAC / Bhuvan / NRSC

MOSDAC ([\[mosdac.gov.in\]](#)): **SSO registration required, no separate API key** — download the Python [\[mdapi\]](#) client ([\[mosdac.gov.in/software/mdapi.zip\]](#)), put username/password + [\[datasetId\]](#) in [\[config.json\]](#), run (needs [\[requests\]](#)). Find the exact [\[datasetId\]](#) (= Product Name) at [\[mosdac.gov.in/catalog/satellite.php\]](#). Oceansat-3 (EOS-06) products confirmed: OCM-3 chlorophyll/AOD/Kd (e.g. [E060CM_L2C_AD](#)), SCAT-3 winds; **HDF5/NetCDF**.

General users get **L1 with 3-day latency**; L2+ nearer real-time. Remember the SSTM fault — no operational Oceansat-3 SST.

Bhuvan / NRSC — WMS/WFS geospatial layers (incl. a PFZ layer mirror) via `bhuvan-app1.nrsc.gov.in`;

Bhoonidhi (`bhoonidhi.nrsc.gov.in`) for EO product download. Good for base map layers and bathymetry overlays.

NIOT — in-situ buoys (ground truth for your reliability layer)

OMNI buoy network: exactly **12 deep-sea OMNI buoys (7 in Bay of Bengal, 5 in Arabian Sea), plus 4 coastal buoys and 2 tsunami buoys** (NIOT OMNI-RAMA portal). They transmit **hourly real-time** surface met + subsurface (T/S/currents to 500 m) + waves (selected buoys). NIOT deploys/maintains; **INCOIS manages the data**. Open-data policy (announced 2018) for buoys outside the EEZ via the **OMNI-RAMA joint portal**. These hourly observations are exactly what you backtest forecasts against for the Reliability Horizon.

CMLRE / NCCR

CMLRE (deep-sea/biodiversity, e.g. FORV Sagar Sampada cruise data) and NCCR (coastal water quality/pollution) hold valuable but **mostly non-API, report/portal-based** data. Realistic hackathon use: cite them as authoritative context for the causal/ecosystem agent (e.g. bycatch, water-quality) and ingest any downloadable datasets manually; don't promise live integration.

International fallbacks (essential for resilience and for waves)

Copernicus Marine (CMEMS) — free account, no key; `pip install copernicusmarine`; `copernicusmarine login` then `subset`. Verified dataset IDs covering the Indian Ocean/BoB/Arabian Sea:

Waves: `cmems_mod_glo_wav_anfc_0.083deg_PT3H-i` (VHM0 significant wave height, VTPK peak period, VMDB direction, swell components; 1/12°, 3-hourly, 10-day) — **your reliable wave source**.

Currents/physics: `cmems_mod_glo_phy-cur_anfc_0.083deg_PID-m` (uo/vo daily), plus thetao/so variants.

Chlorophyll: `cmems_mod_glo_bgc-pft_anfc_0.25deg_PID-m` (chl, 0.25°, daily).

NetCDF/Zarr output; **dataset IDs/versions rotate ~biannually — run `copernicusmarine describe` before the finale**.

NASA OceanColor / PODAAC / Earthdata (MODIS-Aqua, VIIRS, GHRSST SST — Earthdata login), **NOAA ERDDAP / CoastWatch, NOAA Tides & Currents API** (`api.tidesandcurrents.noaa.gov`), **GEBCO** bathymetry grid, **Open-Meteo Marine API** (no key — fastest wave/weather fallback for a demo).

Boundaries/MPA/AIS: marineregions.org (EEZ v12, WFS at `geo.vliz.be/geoserver`, CC-BY), **Protected Planet / WDPA** (MPA polygons, monthly updates), **Global Fishing Watch** (AIS/effort + curated reference layers), **OpenStreetMap** coastline. For the India-Sri Lanka IMBL, use the agreed 1974/76 boundary geometry rather than a computed median.

Ingestion classification (what a team can wire up in days)

Source	Access type	Auth	Format	Realistic to wire in days?
INCOIS ERDDAP	OPeNDAP/REST (griddap/tabledap)	none	NetCDF/CSV/JSON	Yes — start here
IMD api.imd.gov.in	REST/JSON + GeoJSON	none	JSON	Yes
CMEMS	Python toolbox	free acct	NetCDF/Zarr	Yes
Open-Meteo Marine	REST/JSON	none	JSON	Yes (demo fallback)
marineregions/WDPA/GEBCO	bulk download (shp/geojson/tif)	none	vector/raster	Yes (load once into PostGIS)
INCOIS PFZ/OSF	WMS + text scrape	none	WMS/text/PDF	Partial — parse text
MOSDAC	Python <code>mdapi</code> client	SSO reg	HDF5/NetCDF	Partial — registration + latency
NIOT OMNI buoys	portal (INCOIS-managed)	portal	NetCDF/tabular	Partial — for backtesting
CMLRE/NCCR	reports/portal	varies	PDF/tabular	Manual only

Resilient-ingestion design: adapters emit a common record `{variable, value, lat, lon, valid time, source, issued_time, unit}`. Use **xarray + dask** for NetCDF/GRIB; persist subsetted fields as **Zarr / Cloud-Optimized GeoTIFF**; precompute per-coastal-node time series into **PostGIS + TimescaleDB**. Cache by natural cadence (PFZ 3×/week; OSF every 12 h; IMD bulletins a few times/day; CMEMS wave update every 12 h). **Degrade gracefully:** INCOIS SST down → NOAA_AVHRR_AMSR → CMEMS thetao; MOSDAC chlorophyll stale → CMEMS BGC chl; INCOIS waves unavailable → CMEMS VHM0 → Open-Meteo. **Always surface data provenance + age;** refuse/abstain rather than serve stale safety advice.

3. Genuinely Novel, Non-Obvious Differentiators

Every competing team will ship "multi-agent + RAG + map." Below, each idea is scored on **Novelty (vs Indian incumbents)**, **Feasibility (hackathon)**, and **Demo/Jury appeal**. Incumbents to differentiate against: **INCOIS SAMUDRA/SAMUDRA 2.0** (tsunami/high-wave/swell alerts, PFZ, 5-day OSF, tides, 10 languages), **GEMINI** (GAGAN/NavIC deep-sea alerts) + **DAT-SG** distress transmitter + **Sagarmitra, mKRISHI Fisheries** (TCS+CMFRI+INCOIS) and **Fisher Friend** (MSSRF), and the research prototype **Jal Anveshak** (fine-tuned LLaMA-2 for PFZ navigation).

Tier 1 — build these; they win the round

Advisory Reliability Horizon, done rigorously (flagship). Learn, per region and per forecast lead-time, how skillful past advisories were by **backtesting forecasts against NIOT OMNI buoy observations**, using **CRPS, Brier score and reliability diagrams**, and wrap point/interval forecasts in **conformal prediction** for distribution-free coverage guarantees. Then present every answer with a *confidence that decays with horizon*:

"Tomorrow-morning wave forecast here has been within ± 0.4 m 90% of the time at 24 h lead; at 5-day lead, only 60%." **Novelty: very high** — no Indian fisher system quantifies its own trust. **Feasibility: medium** (needs historical forecast+obs pairs; have a synthetic-hindcast fallback). **Demo/jury: very high** — this is the "wow, it knows what it doesn't know" moment. (Grounded in 2024–2026 literature applying conformal prediction to AI weather forecasts and tropical-cyclone tracks.)

Boat-capability-relative risk framing. The same 2.5 m sea state is routine for a mechanized trawler and lethal for an FRP *vallam*/catamaran. Personalize the Safety Score by **vessel class, engine power, range, freeboard and safety envelope**, stored in agent memory. **Novelty: high** (incumbents use generic colour-coded sea-state bands). **Feasibility: high. Demo/jury: high** — visibly humane and technically specific.

IMBL-crossing prevention with predictive drift. Real-time **distance-to-boundary** + a forward projection using the vessel's heading/speed and forecast current, warning *before* the crossing. Directly addresses the arrests (346 fishermen / 44 trawlers in 2025). **Novelty: high** (GEMINI warns of weather, not boundary proximity).

Feasibility: high with PostGIS. **Demo/jury: very high** — a tangible life-and-liberty impact.

Causal fish-productivity engine. For "why has productivity declined here?", run a constrained **causal DAG / Granger-causality** analysis over co-located time series — SST anomaly, chlorophyll trend, upwelling index, **marine-heatwave** days, freshwater flux, and fishing effort (from Global Fishing Watch) — and return an *evidence-ranked hypothesis set*, not a single guess. Ground it in documented Arabian-Sea science (warming → stratification → shift from diatoms to *Noctiluca* → altered productivity; rising marine-heatwave days). **Novelty: high. Feasibility: medium. Demo/jury: high** (researcher persona loves it).

Deterministic replay + provenance graph. Every agent decision is logged as an auditable graph (dataset → timestamp → agent → formula → output) and can be **replayed deterministically**. For a *government safety* system this is the strongest trust signal you can give a jury, and it cleanly replaces the vague "7-point integrity check." **Novelty: high** (agentic-AI-specific). **Feasibility: high. Demo/jury: very high.**

Tier 2 — strong additions if time allows

6. **Participatory calibration loop:** fishermen report actual catch/sea-state; the system uses it to recalibrate PFZ confidence and safety thresholds locally (a lightweight active-learning loop). Novel; feasibility medium.
7. **Multi-objective Pareto zone/route ranking:** optimize *catch-probability vs fuel vs risk vs IMBL/MPA compliance* and show the trade-off frontier, not one answer. Novel framing; feasible.
8. **Anomaly detection module:** HAB (*Noctiluca*), oil-spill and **marine-heatwave** flags from ocean-colour + SST anomalies, tied into INCOIS ABIS. Scientifically anchored; feasible as a classifier/threshold demo.
9. **Sustainability constraints as first-class:** juvenile/spawning seasons, state fishing-ban periods, MPA no-take zones and bycatch appear as *hard constraints* in the reasoning solver. Cheap, high jury appeal (aligns with the "sustainability" scoring dimension).
10. **Agentic-AI innovations:** an **adversarial verifier/critique agent**, **tool-use planning with cost budgets**, **evidence-sufficiency gating with abstention**, and **dynamic DAG re-planning**. These make the "agentic" claim real rather than cosmetic.

Do NOT reinvent (integrate instead): delay-tolerant/offline delivery already exists as **GEMINI (GAGAN/NavIC)**, **DAT-SG distress** and **Sagarmitra**; SMS/WhatsApp/Telegram advisory reach already exists via INCOIS. Position ORCA's offline mode as *complementary* (cache last-known advisory + on-device small model + hand-off to GEMINI/DAT-SG for the truly-offshore leg), and say so explicitly.

Honest verdict on ORCA's five claimed USPs:

(1) Causal/Hypothesis Reasoning — genuinely differentiating *if* grounded in real drivers + a DAG/Granger method; as a bare LLM prompt it is not. **Sharpen with method + data.**

(2) What-If Marine Simulator — appealing but easy to over-claim. Scope it to *forecast-driven scenario replay* ("if I leave 3 h later / go 10 km further"), not a physics ocean model. **Keep, but scope honestly.**

(3) Context-Aware Marine Geofencing — strong and real; make the IMBL predictive-drift the hero. **Keep and elevate.**

(4) Advisory Reliability Horizon — your best USP; currently under-built. Add conformal prediction + buoy backtesting. **Invest most here.**

(5) Mission Planner — good product framing (goal → constraint-aware plan with departure/fishing-window/return). Novel as an *agentic constraint solver*; make sure it uses the Pareto trade-off and geofence/sustainability constraints. **Keep, connect to #7 and #9.**

4. End-to-End Solution Flow / Build Blueprint

Request lifecycle (single golden path):

Input (text or voice, any Indian language) → **language detection + ASR** (Bhashini/IndicWhisper) → normalize to a canonical intent request.

Intent + entity extraction (LLM with a strict JSON schema): {intent, location (resolved to lat/lon), time window, objective, constraints, vessel_profile}. Ask a clarifying question if a required slot is missing (multi-turn).

Context/memory resolution: load vessel profile (class, power, range), home port, past queries, language preference.

Planner agent emits a **dynamic task DAG** with tool/agent selection under a **cost budget** (token + latency + API-call caps).

Parallel agent execution (fan-out), each agent calling only its registered tools.

Evidence retrieval: structured (PostGIS/TimescaleDB) + unstructured (**pgvector** RAG over advisory text, SOPs, bulletins).

Cross-agent evidence fusion + conflict resolution (e.g. INCOIS vs CMEMS wave disagreement → prefer higher-reliability-scored source, widen uncertainty).

Verifier/critique agent + evidence-sufficiency gating: if evidence is stale/insufficient/conflicting beyond a threshold → **abstain** ("I can't safely answer; nearest reliable data is 6 h old").

Decision/scoring kernel computes Safety Score, Fishing-Zone Intelligence, Route, Reliability Horizon.

Explanation + provenance generation (which sources, ages, formulas).

Multilingual response synthesis (IndicTrans2 + Indic-Parler-TTS) + **map/chart** (MapLibre GL + Shapely/GeoPandas).

Proactive alert subsystem runs independently on a schedule, evaluating geofences and thresholds for at-sea users and pushing warnings.

Recommended agent roster & responsibilities:

Agent	Responsibility	Key tools
Planner/Orchestrator	intent→DAG, tool selection, budget	LLM + registry
Marine Data	SST/chl/PFZ/currents retrieval	INCOIS ERDDAP, MOSDAC, CMEMS
Weather Intelligence	wind/waves/cyclone/lightning	IMD JSON, CMEMS waves, Open-Meteo
Geospatial/Geofence	IMBL/EEZ/MPA distance & containment	PostGIS
Ecosystem/Causal	productivity, HAB, causal DAG	time-series + Granger
Risk/Safety	boat-relative safety score	scoring kernel
Route	isochrone-A* over cost surface	GeoPandas/NetworkX
Reliability	CRPS/Brier + conformal intervals	buoy hindcast store
Verifier/Critique	conflict + sufficiency gating	rules + LLM
Response Synthesizer	multilingual NL + provenance	Bhashini

Inter-agent message schema (suggested): [{task_id, agent, status, inputs, evidence[], result, confidence, sources[], issued_time, cost_used}] on a shared typed state object (Pydantic).

Orchestration recommendation: use **LangGraph** for the stateful agent graph (checkpointing, memory, streaming, conditional branching, human-in-loop) — its DAG/state-machine model fits "dynamic DAG re-planning" well and is well-supported — **but keep deterministic tool execution in FastAPI microservices** so runs are reproducible and replayable. Rationale: LangGraph gives you the agentic control flow cheaply; isolating computation in deterministic services gives you the auditability/replay that a safety jury rewards. (If LangGraph's abstractions slow you down, a custom FastAPI DAG executor is an acceptable fallback — several 2025–2026 systems report LangGraph churn.)

Model-choice discipline (state this explicitly to the jury):

LLM only for: language detection, intent/entity extraction, evidence fusion narrative, explanation synthesis, critique.

Classical ML / statistics for: reliability scoring (CRPS/Brier/conformal), causal discovery (Granger/DAG), anomaly detection.

Deterministic rules / PostGIS / graph search for: safety thresholds, geofencing, routing. *Never let the LLM invent a safety number* — it only explains numbers the kernel computed.

Decision kernels (concrete):

Marine Safety Score = weighted combination of significant wave height (VHM0), wind speed, swell period and squall/lightning risk, each normalized against the **vessel-class thresholds**, then multiplied by the **reliability factor** for that region/lead-time; output a 0–100 score *plus* a confidence band; **hard-abstain** if any driver is stale beyond its cadence. Threshold bands per boat class (e.g. FRP < 1.5 m Hs "caution," < 2 m "avoid"; trawler higher). **Fishing-Zone Intelligence** = Pareto rank over {PFZ proximity/age, SST-front strength, chlorophyll gradient, distance/fuel cost, safety score, IMBL/MPA/ban compliance}. Return the trade-off frontier, not one point.

Reliability Horizon = per (region, variable, lead-time) skill learned by backtesting forecasts vs OMNI buoy obs (CRPS/Brier); conformal interval width grows with lead-time → the visible "confidence decay curve."

Route optimization: recommend **isochrone-A*** (or A* over a gridded cost surface) where cell cost $= f(VHM0, \text{current vector projected on heading, wind, bathymetry/depth})$ with **hard constraints** from geofence polygons (IMBL buffer, MPA no-take, ban zones) removing/penalizing cells. North-Indian-Ocean precedent exists (Sen & Padhy 2015, *Applied Ocean Research*). [Cambridge Core](#)

↗
Isochrone-A* and Isochrone-Dijkstra outperform naive great-circle in published weather-routing benchmarks.

Geofencing (PostGIS): load EEZ (marineregions v12), the **agreed India-Sri Lanka IMBL geometry**, and WDPA MPA polygons. Use `ST_Contains`/`ST_Intersects` for inside-restricted checks; `ST_Distance` on `geography` for true metric distance-to-IMBL; `ST_Buffer` to build graded warning bands (e.g. 5 km "amber", 2 km "red"); project future position from heading+current for predictive alerts.

Proactive alert subsystem: a scheduled worker that, for each active at-sea user, evaluates (a) IMD cyclone/wind GeoJSON containment, (b) high-wave/lightning thresholds, (c) geofence proximity, and pushes CAP-formatted alerts (WebSocket/FCM in-app; document GEMINI/DAT-SG hand-off for offshore).

Caching/precomputation: precompute coastal-node time series nightly; tile map layers; memoize agent results keyed by (intent, location-cell, valid-hour). **Offline/degraded mode:** cache last-known advisory + a small on-device model for cached-data Q&A; explicit "data age" banner.

Evaluation strategy (build this — it scores on feasibility & rigor):

Golden query set (~30 queries incl. the five canonical ones) with expected behaviours.

Hindcast backtesting of the reliability layer against OMNI buoy records.

Ablation (with/without reliability, with/without verifier) to show each component's value.

Abstention/hallucination test: feed stale/missing data and confirm the system refuses.

Realistic build sequence:

Days 1–2: INCOIS ERDDAP + IMD JSON + CMEMS waves + Open-Meteo fallback; load EEZ/IMBL/MPA into PostGIS; prove one golden query end-to-end.

Days 3–5: Planner DAG + 4–5 core agents; safety & fishing scores; distance-to-IMBL; proactive alert loop; MapLibre UI.

Days 6–8: Reliability horizon (buoy backtest), causal engine, Bhashini voice, provenance/replay panel, verifier/abstention; demo polish + persona views.

Demo script (highest-impact path): a fisherman asks in **Tamil, by voice**, "Is it safe to go tomorrow morning, and where should I fish?" → planner decomposes → agents fetch INCOIS/IMD/CMEMS in parallel → **boat-relative**

Safety Score = 72/100 with a confidence band that decays past 3 days → **geofence warning: "You would be 4 km from the Sri Lankan maritime boundary — stay west"** → **Pareto fishing recommendation** (higher catch-probability zone vs a safer/cheaper one) → **provenance panel** showing each source and its age → spoken Tamil answer. Then trigger a **proactive cyclone alert** to show push, and open the **researcher view** on "why has productivity declined here?" to show the causal DAG. Finish on the **deterministic replay** of the whole decision.

Highest-risk assumptions to validate on Day 1: (1) MOSDAC registration/latency — if blocked, rely on CMEMS+INCOIS; (2) INCOIS wave/OSF griddap ID — confirm live or use CMEMS VHM0; (3) OMNI buoy historical access for backtesting — if unavailable, ship reliability as a documented design with a synthetic hindcast demo; (4) Bhashini API quotas — cache and pre-generate demo audio; (5) IMBL geometry correctness — use the agreed 1974/76 boundary, not a median line.

Failure-safe behaviour: the verifier + evidence-sufficiency gate must make ORCA **abstain and state why** ("data is 8 h old; I won't give a safety verdict") rather than hallucinate. For a life-safety, government-facing system, *calibrated refusal is a feature*, and demonstrating it will impress a domain jury more than a confident wrong answer.

Recommendations (staged, with triggers)

Lock the positioning first: "ORCA is the agentic reasoning + trust layer over INCOIS/IMD/ISRO feeds," not a new advisory source. This single framing neutralizes the biggest jury objection.

Invest most engineering in the Reliability Horizon + boat-relative risk + IMBL predictive geofence — these three are your defensible, non-duplicative core. Build them before polishing the chat UI.

Ship the JSON/GeoJSON sources first (INCOIS ERDDAP, IMD, CMEMS, Open-Meteo, PostGIS polygons); treat MOSDAC/NIOT as enhancements with fallbacks.

Add the four persona views and CAP-formatted proactive alerts to score directly on the disaster-management theme and multi-stakeholder clause.

Trigger to change plan: if buoy hindcast data can't be assembled by Day 4, present reliability as a rigorous *design* with a synthetic demo rather than dropping it — the concept still scores on novelty. If MOSDAC access is blocked, do not mention Oceansat-3 SST at all; lead with OCM-3 chlorophyll + SCAT winds or CMEMS.

Rehearse the abstention demo — deliberately show ORCA refusing on stale data; it is a differentiator, not a weakness.

Caveats

INCOIS restricts some in-EEZ data; PFZ has **no clean JSON API** (text/WMS only); the INCOIS wave/OSF griddap ID needs live confirmation. MOSDAC requires registration and imposes a **3-day L1 latency** for general users, and **Oceansat-3's SSTM is non-operational** (no operational Oceansat-3 SST). CMEMS dataset IDs/versions **rotate ~biannually** — verify with [copernicusmarine describe](#) before the finale. Conformal/skill scoring needs historical forecast+observation pairs (OMNI buoys) that may be hard to assemble in hackathon time — keep a synthetic-hindcast fallback. The **IMBL must use the agreed 1974/76 India-Sri Lanka boundary**, not a computed median, to avoid dangerously wrong warnings. Fisher-arrest figures (346 fishermen / 44 trawlers in 2025) are from Sri Lankan Navy statements via Indian press and should be cited as such. Several existing systems (SAMUDRA, GEMINI, DAT-SG, mKRISHI, Fisher Friend, Jal Anveshak) already cover advisory delivery, offline alerting and PFZ navigation — ORCA's defensibility rests entirely on the reasoning/trust/causal/geofencing layer, so keep that front and centre.